Smart Data Exploration Hub:

An AI-Powered Chatbot Application

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Project outreach page: Here

Key Terms

- **SPEAR** - (Seamless System for Prediction and EArth System Research)
  - NOAA GFDL's Earth system model for seasonal to long-scale climate prediction and projection.
- **LLMs** - (Large Language Models)
  - A neural network trained in large text datasets to generate and interpret human language.
- **MCP** - (Model Context Protocol)
  - A standardized interface enabling LLMs to securely interact with data and tools.
- **RAG** - (Retrieval-Augmented Generation)
  - A technique that grounds AI responses using retrieved documents.
- **STAC** - (SpatioTemporal Asset Catalog)
  - A standardized metadata framework for indexing and accessing geospatial datasets.

The Why, What and How



Why

To expand access to SPEAR large-ensemble earth system data and insights.



What

SPEAR Earth System Data Assistant, publicly accessible with grounded responses from supplied documentation.



How

Streamlit,
Large Language Models,
Model Context Protocol,
Retrieval-Augmented Generation.

Providing Insight and Lowering Barriers to SPEAR Results

Goals:

- Making earth system model data more reachable.
  - For those with AND without climate/science backgrounds.
- Continuous user engagement and enhanced service delivery.
- Building best practices for others with similar interests.
  - Publicly available on Github with clear documentation.[5]
- To provide a conversational method of inspecting, extracting and analyzing SPEAR output.



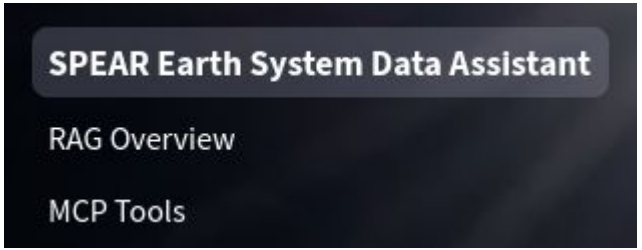
Grounded Responses Using Supplied Documentation

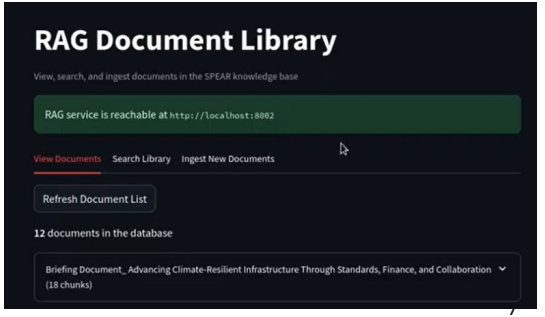
- Credible and reproducible AI responses are paramount.
- Explicit instructions for increased safety.
  - Restricted discussion topics.
  - Use of responsible AI tools.
- Responses based on provided documentation and training base knowledge.
- Qualitative confidence estimates for responses.
- Future steps include pre-computed diagnostic plots.



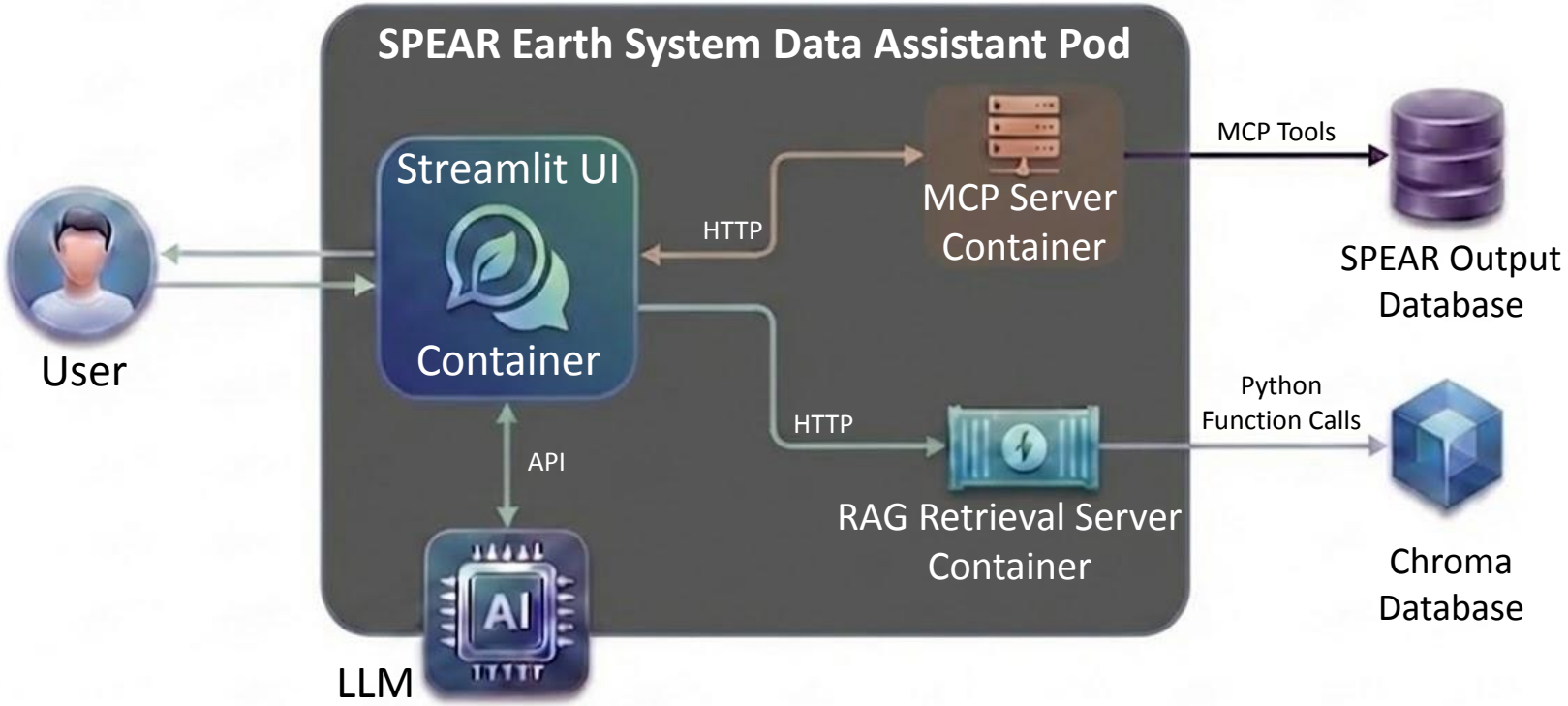
Increased Transparency in AI Responses

- Users are informed about the sources of content used in responses.
- Ability to view and read the documents in the chatbot supplemental knowledge base.
- Interactive overview of data extraction tools and files used to generate tables, plots or analysis.
- Ability to investigate sources with two additional tabs:
  - RAG Overview
  - MCP Tools





SPEAR Earth System Data Assistant



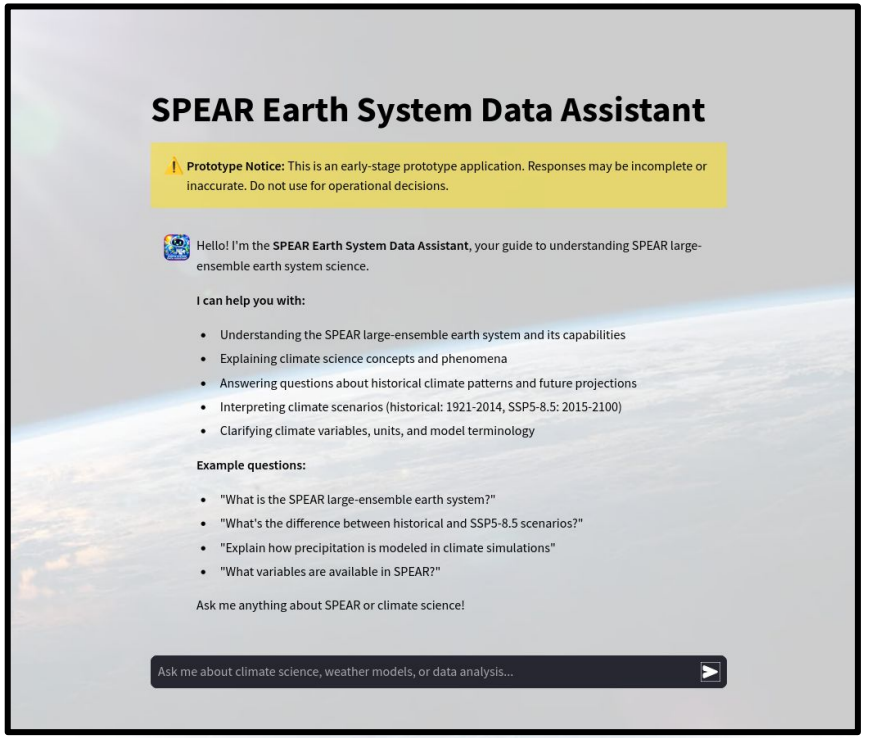
Streamlit UI

A web application framework designed to easily build interactive dashboards directly from Python scripts (no HTML or Java needed).

Serves as our chatbot UI displayed via localhost.

Allows for:

- **Topic Regulation**, **Controlled Vocabulary, Multi-User Support and Authentication**
- Scripted welcome message.
- Qualitative confidence estimates provided at the end of each response.
- Easily view chatbot logs and conversation history.



Large Language Model (LLM)

- Compatible with locally and remotely hosted models.
- Local Ollama
  - Llama, Qwen, Gemma
- Remote API
  - Gemini, Claude
- Handles response generation and tool execution.
- Decides between RAG content or MCP tool usage.

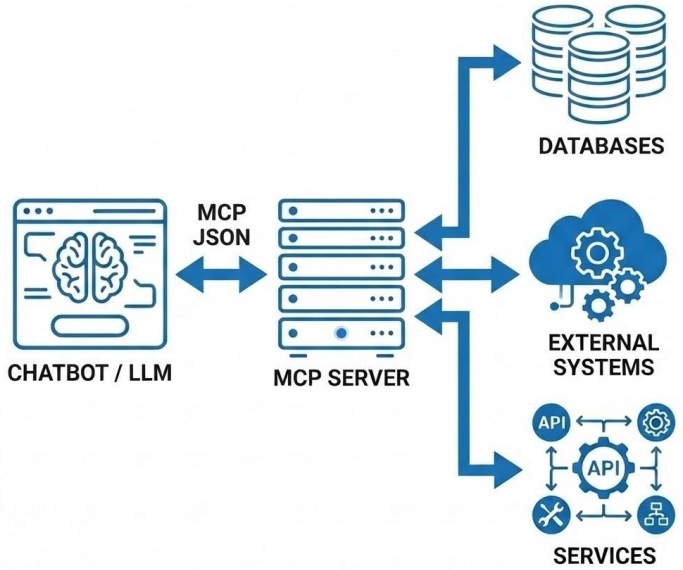


Model Context Protocol (MCP)

A standardized communication method that enables Large Language Models (LLMs) to exchange information with external systems, databases and services.

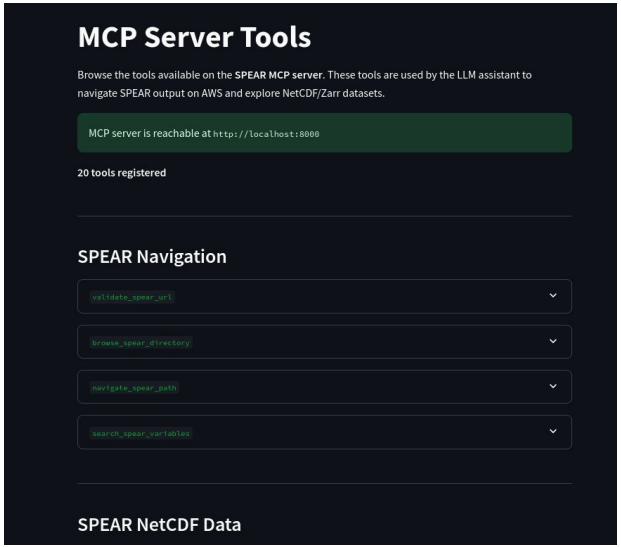
Provides secure and structured access to data and tools that would otherwise be unavailable to AI models.

Uses a simple JSON format for consistent interoperable communication (HTTP).



The SPEAR Earth System Data Assistant MCP Server

- Tools for NetCDF inspection and extraction
  - Metadata
  - Data arrays
  - Directory and file browsing
- 3 different server configurations
  - AWS cloud
  - Locally mounted filesystems
  - SpatioTemporal Asset Catalogs (STAC)



Retrieval-Augmented Generation (RAG)

- Retrieves info from trusted documents before responding.
- Ensures answers are grounded in SPEAR and stakeholder documentation, such as ASCE.
- Reduces hallucinations and improves factual accuracy.
- Allows the chatbot to answer questions using[13] up-to-date, project-specific information.
- 3 primary stages to RAG.

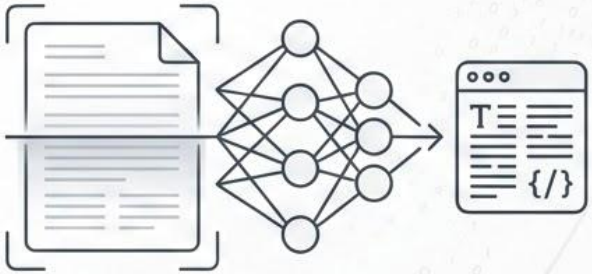




RAG: Stage 1

Text Extraction from PDF

Primary Method: Nougat



- A neural OCR model converting from pixels to words.
- Understands document layouts and produces structured text and equations.

Secondary Method: pypdf



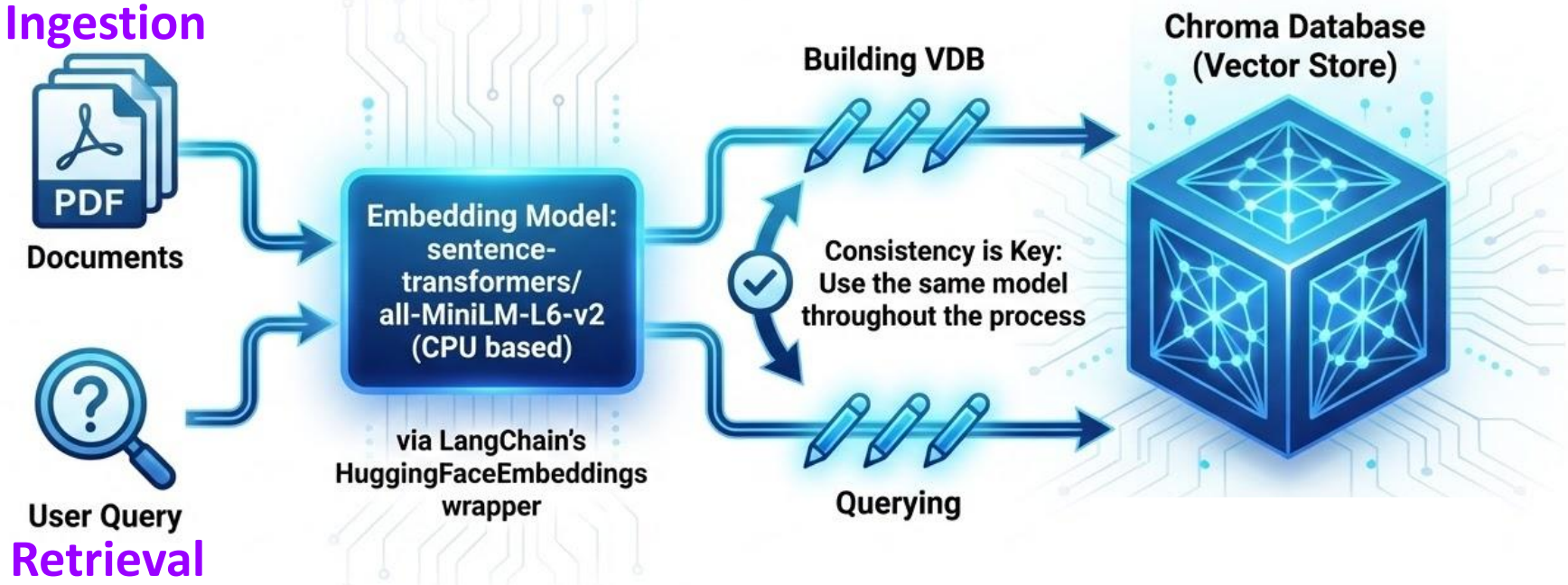
Extracts embedded PDF text. Only used on pages skipped by Nougat.



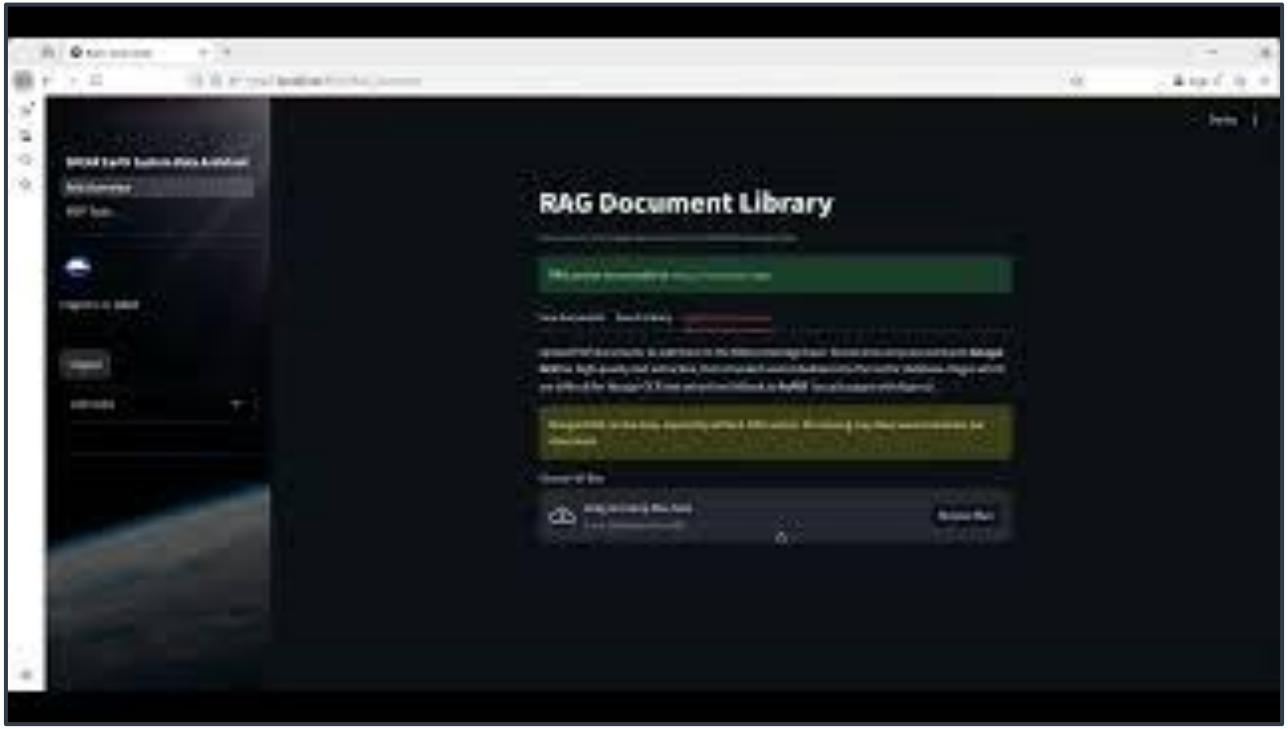
Prevents “repetition collapse” on figure-dense pages by skipping problematic pages.

RAG: Stages 2 and 3

Ingestion and Retrieval from Chroma Database



SPEAR Earth System Data Assistant: Live Demo



*If video is blurry, set playback quality to 1080p^{HD}



On the Horizon

- Reduce computational load.
  - Pre-computed diagnostic plots.
- Optimize data interactions.
  - Zarr and Icechunk, instead of NetCDF.
- Fine-tuning of a locally run LLM.
  - Benchmarking included.
    - Figure generation enhancements (manuscript ready plots).

Github Repositories

- Streamlit UI:
  - https://github.com/zappalaja/spear-climate-chatbot
- MCP Server:
  - https://github.com/zappalaja/spear-mcp-test
- RAG Ingestion:
  - https://github.com/zappalaja/SPEAR-RAG-Ingestion
- RAG Retrieval Service:
  - https://github.com/zappalaja/SPEAR_RAG_Retrieval_Service
- **Full SPEAR ESDA** (all components above):
  - https://github.com/zappalaja/SPEAR_Earth_System_Data_Assistant
- SPEAR AWS Output
  - https://noaa-gfdl-spear-large-ensembles-pds.s3.amazonaws.com/index.html#SPEAR/GFDL-LARGE-ENSEMBLES/CMIP/NOAA-GFDL/GFDL-SPEAR-MED/

Questions?

Submit questions/feedback anytime: https://bit.ly/4lF1rT1

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